3Consulting
BUILD TRUST IN THE DIGITAL ECONOMY

ENTERPRISE DATA GOVERNANCE · CASE STUDY

# SafeData: Automated NDPA & GDPR Consent Disposal

BI Scientist • AI/ML Engineer • Python Developer

Enterprise retention platform enforcing automated PII disposal, 30-day quarantine, and SHA-256 hash-chain audit logging across distributed client databases.

THE COMPLIANCE CHALLENGE

# Why Enterprise Data Retention Breaks

NDPA Section 34 and GDPR Article 17 require verifiable, irreversible destruction of customer data upon consent withdrawal.

## Orphaned PII in Backups

User consent is withdrawn in CRM, but records persist in analytics replicas and backups.

No automated system maps customer PII across disparate microservices.

High risk of multi-million naira NDPA enforcement penalties.

## Manual Spreadsheet Failure

DPOs rely on spreadsheets to track departmental data retention schedules.

Zero dual-approval or legal-hold safeguards before accidental deletion.

Audit trails are unverified and fail regulatory inspection.

## Unprovable Deletion

Regulators demand immutable proof that deleted records cannot be recovered.

Standard DB DELETE logs lack cryptographic non-repudiation.

SafeData creates mathematical disposal certainty.

SYSTEM ARCHITECTURE

# Automated Scanners, Retention Engine & Hash-Chain Audit

End-to-end data lifecycle orchestration deployed on hardened FastAPI / PostgreSQL multi-tenant infrastructure.

## Intake & Webhooks

- Scoped API keys & HMAC signatures.
- Listens for consent-withdrawn events.
- Maps PII identifiers across client tables.

## Retention Engine

- Jurisdiction-aware retention schedules.
- 30-day quarantine isolation buffer.
- Dual-approval workflows & legal holds.

## Tamper-Proof Audit

- SHA-256 chained disposal receipts.
- Cryptographic proof for NDPC filings.
- Zero data recovery post-purge.

OPERATIONAL REALITY

# SafeData Governance Console

Live DPO console tracking disposal queues, RoPA registers, and legal-hold approvals.

- **Live Disposal Pipeline:** Visual progress of records transitioning from Active → Quarantine → Approved → Purged.
- **Dual-Key Approvals:** Legal & Compliance officers must co-sign high-sensitivity dataset deletions.
- **Automated RoPA Registers:** Dynamic generation of Record of Processing Activities for NDPA audits.

FastAPI | PostgreSQL | Docker | NDPA / GDPR



safedata.3consult-ng.com/app/retention

ENTERPRISE IMPACT

# Quantified Compliance Results

Production governance metrics delivered across multi-tenant enterprise deployments.

| Audit Trail Integrity | Orphaned Records | Quarantine SLA | Audit Time Saved |
|---|---|---|---|
| 100% | 0 | 30 Days | 85% |
| SHA-256 chained hashes | Zero untracked PII leaks | Standard cooldown buffer | Automated DPO reporting |

## Regulatory Readiness

- Full compliance with Nigeria Data Protection Act (NDPA 2023).
- GDPR Article 17 (Right to Erasure) verified audit readiness.
- Immutable non-repudiation logs ready for regulatory submission.

## Enterprise Reliability

- Zero data loss on legal-hold protected litigation records.
- Multi-tenant PostgreSQL isolation with tenant-level key rotation.
- Asynchronous background workers scan 1M+ rows without DB locks.

CASE STUDY TAKEAWAY

# Keep what you must. Release what you should.

SafeData transforms regulatory compliance from a chaotic legal panic into an automated, mathematically auditable engineering asset.

**Company:** 3Consulting Limited (Lagos, Nigeria)
**Live Platform:** safedata.3consult-ng.com